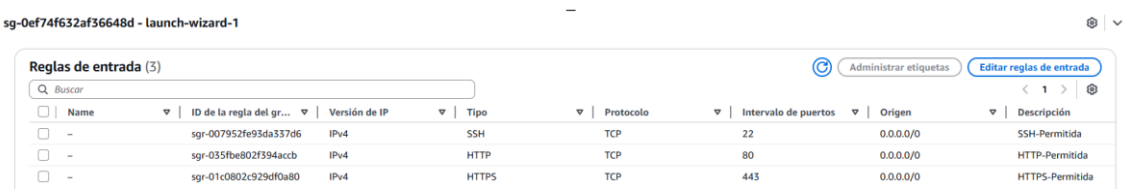
En nuestra maquina de aws y añadiendo una regla https accederemos a nuestra ip elástica por ssh




```
C:\Users\UsuarioASIR\Desktop\PEM>ssh -i .\labsuser.pem ubuntu@54.84.84.203
```

Instalación de certbot

```
ubuntu@akatsuki:~$ sudo apt-get install certbot python3-certbot-apache
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
certbot is already the newest version (2.9.0-1).
The following additional packages will be installed:
  augeas-lenses libaugeas0 python3-augeas
Suggested packages:
  augeas-doc augeas-tools python-certbot-apache-doc
The following NEW packages will be installed:
  augeas-lenses libaugeas0 python3-augeas python3-certbot-apache
0 upgraded, 4 newly installed, 0 to remove and 56 not upgraded.
Need to get 626 kB of archives.
After this operation, 3276 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 augeas-lenses all 1.14.1-1build2 [323 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 libaugeas0 amd64 1.14.1-1build2 [166 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-augeas all 0.5.0-1.1 [9124 B]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-certbot-apache all 2.9.0-1 [128 kB]
Fetched 626 kB in 0s (22.4 MB/s)
Selecting previously unselected package augeas-lenses.
(Reading database ... 72886 files and directories currently installed.)
Preparing to unpack .../augeas-lenses_1.14.1-1build2_all.deb ...
Unpacking augeas-lenses (1.14.1-1build2) ...
Selecting previously unselected package libaugeas0:amd64.
Preparing to unpack .../libaugeas0_1.14.1-1build2_amd64.deb ...
Unpacking libaugeas0:amd64 (1.14.1-1build2) ...
Selecting previously unselected package python3-augeas.
Preparing to unpack .../python3-augeas_0.5.0-1.1_all.deb ...
Unpacking python3-augeas (0.5.0-1.1) ...
Selecting previously unselected package python3-certbot-apache.
Preparing to unpack .../python3-certbot-apache_2.9.0-1_all.deb ...
Unpacking python3-certbot-apache (2.9.0-1) ...
Setting up augeas-lenses (1.14.1-1build2) ...
Setting up libaugeas0:amd64 (1.14.1-1build2) ...
Setting up python3-augeas (0.5.0-1.1) ...
Setting up python3-certbot-apache (2.9.0-1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.5) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.
```

Cómo usar certbot

Estos comandos solicitarán un certificado SSL/TLS para tu dominio especificado y configurarán automáticamente el servidor web para usarlo.

```
ubuntu@akatsuki:~$ sudo certbot --apache -d akatsuki.institutodh.net
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Enter email address (used for urgent renewal and security notices)
 (Enter 'c' to cancel): marioreig17@institutodh.net

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Please read the Terms of Service at
https://letsencrypt.org/documents/LE-SA-v1.5-February-24-2025.pdf. You must
agree in order to register with the ACME server. Do you agree?
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
(Y)es/(N)o: y

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Would you be willing, once your first certificate is successfully issued, to
share your email address with the Electronic Frontier Foundation, a founding
partner of the Let's Encrypt project and the non-profit organization that
develops Certbot? We'd like to send you email about our work encrypting the web,
EFF news, campaigns, and ways to support digital freedom.
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
(Y)es/(N)o: n
Account registered.
Requesting a certificate for akatsuki.institutodh.net

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/akatsuki.institutodh.net/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/akatsuki.institutodh.net/privkey.pem
This certificate expires on 2026-02-02.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for akatsuki.institutodh.net to /etc/apache2/sites-available/000-default-le-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://akatsuki.institutodh.net

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

Veremos activo ya el puerto 443 abierto

```
ubuntu@akatsuki:~$ sudo ss -lntp
State          Recv-Q         Send-Q                 Local Address:Port                 Peer Address:Port        Process
LISTEN         0              4096                      127.0.0.54:53                      0.0.0.0:*            users:(("systemd-resolve",pid=332,fd=17))
LISTEN         0              4096                       0.0.0.0:22                        0.0.0.0:*            users:(("sshd",pid=1276,fd=3),("systemd",pid=1,fd=200))
LISTEN         0              4096                  127.0.0.53%lo:53                       0.0.0.0:*            users:(("systemd-resolve",pid=332,fd=15))
LISTEN         0              511                          *:6083                            *:*            users:(("apache2",pid=2746,fd=6),("apache2",pid=2745,fd=6),("apache2",pid=560,fd=6))
LISTEN         0              4096                      [::]:22                           [::]:*            users:(("sshd",pid=1276,fd=4),("systemd",pid=1,fd=201))
LISTEN         0              511                          *:80                              *:*            users:(("apache2",pid=2746,fd=4),("apache2",pid=2745,fd=4),("apache2",pid=560,fd=4))
```

Certbot actualmente NO añade automáticamente una tarea cron al instalar un certificado, debe añadir al cron

```
ubuntu@akatsuki:~$ sudo certbot renew --dry-run
Saving debug log to /var/log/letsencrypt/letsencrypt.log

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Processing /etc/letsencrypt/renewal/akatsuki.institutodh.net.conf
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Account registered.
Simulating renewal of an existing certificate for akatsuki.institutodh.net

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Congratulations, all simulated renewals succeeded:
  /etc/letsencrypt/live/akatsuki.institutodh.net/fullchain.pem (success)
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
ubuntu@akatsuki:~$ sudo certbot renew
Saving debug log to /var/log/letsencrypt/letsencrypt.log

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Processing /etc/letsencrypt/renewal/akatsuki.institutodh.net.conf
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Certificate not yet due for renewal

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
The following certificates are not due for renewal yet:
  /etc/letsencrypt/live/akatsuki.institutodh.net/fullchain.pem expires on 2026-02-02 (skipped)
No renewals were attempted.
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
ubuntu@akatsuki:~$
```

Y aquí vemos como por https estamos en nuestra pagina default de nuestro dominio


akatsuki.institutodh.net
Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
|       `--  ports.conf
|-- mods-enabled
|       |-- *.load
|       `-- *.conf
|-- conf-enabled
|       `-- *.conf
|-- sites-enabled
|       `-- *.conf
apache2.conf is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
ports.conf is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
Configuration files in the mods-enabled/, conf-enabled/ and sites-enabled/ directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
They are activated by symlinking available configuration files from their respective *-available/ counterparts. These should be managed by using our helpers a2enmod, a2dismod, a2ensite, a2dissite, and a2enconf, a2disconf . See their respective man pages for detailed information.